

exp 25.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 25.py (3.13.5)

```
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
X, y = make_circles(
    n_samples=500, noise=0.1, factor=0.5,
    random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

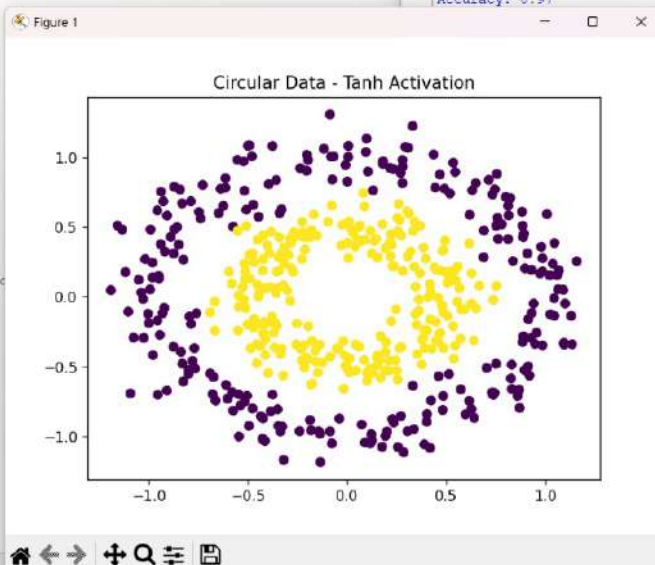
```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="tanh",
    max_iter=1000,
    random_state=42
)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Circular Data - Tanh Activation")
plt.show()
```



MIDLE Shell 3.13.5*

```
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 25.py =====

Warning (from warnings module):
  File "C:/Users/shaik/AppData/Local/Programs/Python/Python313/Lib/site-packages
\sklearn/neural_network/multilayer_perceptron.py", line 785
    warnings.warn(
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and
the optimization hasn't converged yet.
Accuracy: 0.97
```

Ln: 5 Col: 0

Ln: 36 Col: 0

32°C
Mostly sunny



ENG IN 81% 10:53:22 AM 20-08-2026

exp 27.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 27.py (3.13.5)

```
File Edit Format Run Options Window Help
import matplotlib.pyplot as plt
from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_classification(
    n_samples=500, n_features=2, n_redundant=0,
    n_informative=2, n_classes=2, random_state=42
)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    learning_rate_init=0.001,
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

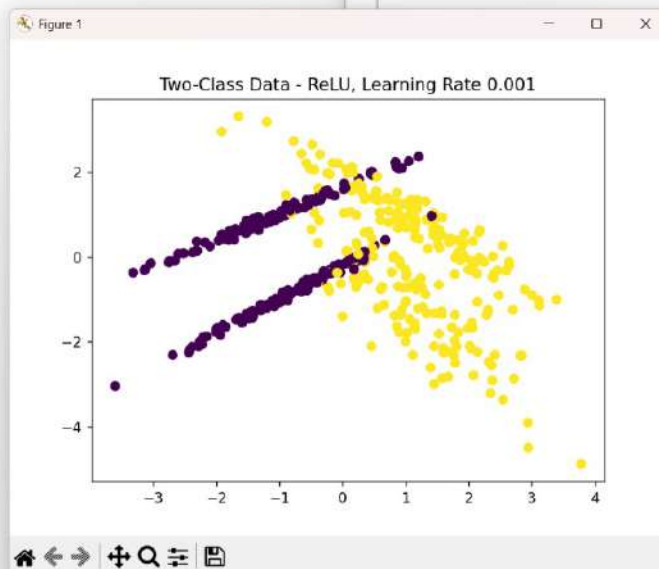
print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Two-Class Data - ReLU, Learning Rate 0.001")
plt.show()
```

IDLE Shell 3.13.5

```
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 27.py =====
Accuracy: 0.89
```



Ln: 5 Col: 0

Ln: 37 Col: 0

```
exp 23.py - C:/Users/shaik/OneDrive/Desktop/MIA0405/exp 23.py (3.13.5)
File Edit Format Run Options Window Help

import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

np.random.seed(42)

n = 500
theta = np.sqrt(np.random.rand(n)) * 4 * np.pi
r = 2 * theta + np.pi

X = np.column_stack((r * np.cos(theta), r * np.sin(theta)))
y = (theta > 2 * np.pi).astype(int)

X += np.random.randn(n, 2) * 0.5

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(20,),
    activation="logistic",
    max_iter=2000,
    random_state=42
)

model.fit(X_train, y_train)
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

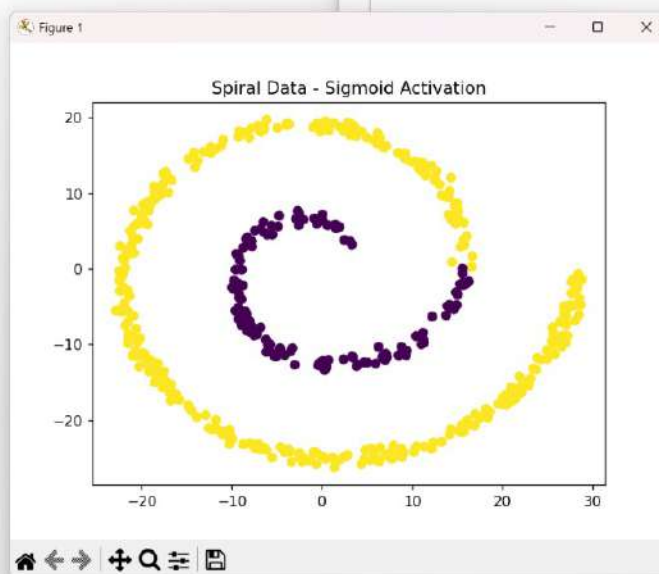
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Spiral Data - Sigmoid Activation")
plt.show()
```

```
"IDLE Shell 3.13.5"
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Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MIA0405/exp 23.py =====
Accuracy: 0.73

Ln: 5 Col: 0
```



exp 29.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 29.py (3.13.5)

File Edit Format Run Options Window Help

```
import matplotlib.pyplot as plt
from sklearn.datasets import make_blobs
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
```

```
X, y = make_blobs(
    n_samples=500, centers=3, n_features=2,
    random_state=42
)
```

```
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
model = MLPClassifier(
    hidden_layer_sizes=(3,),
    activation="tanh",
    max_iter=2000,
    random_state=42
)
```

```
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

```
print("Accuracy:", accuracy_score(y_test, y_pred))
```

```
plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Multi-Class Data - Tanh, 3 Hidden Neurons")
plt.show()
```

"IDLE Shell 3.13.5"

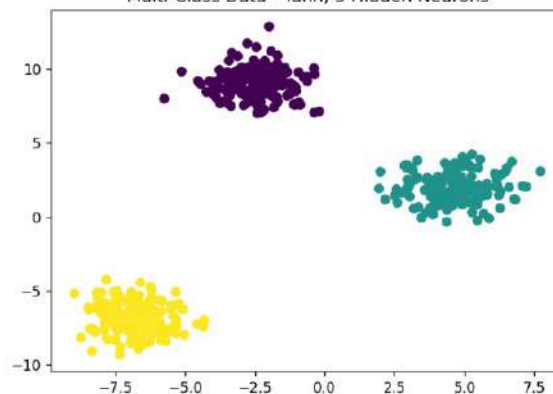
File Edit Shell Debug Options Window Help

Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

```
>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 29.py =====
Accuracy: 1.0
```

Figure 1

Multi-Class Data - Tanh, 3 Hidden Neurons



Navigation icons: Home, Previous, Next, Zoom In, Zoom Out, Full Screen, Save.

Ln: 5 Col: 0

Ln: 36 Col: 0

32°C Mostly sunny



Search



ENG IN



79%

11:03:05 AM 20-08-2026

```
exp 21.py - C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 21.py (3.13.5)
File Edit Format Run Options Window Help
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import make_circles
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

X, y = make_circles(n_samples=500, noise=0.1, factor=0.5, random_state=42)

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = MLPClassifier(
    hidden_layer_sizes=(10,),
    activation="relu",
    max_iter=1000,
    random_state=42
)

model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))

plt.scatter(X[:, 0], X[:, 1], c=y)
plt.title("Circular Data - ReLU Activation")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```

```
"IDLE Shell 3.13.5"
File Edit Shell Debug Options Window Help
Python 3.13.5 (tags/v3.13.5:6cb20a2, Jun 11 2025, 16:15:46) [MSC v.1943 64 bit (AMD64)]
on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/shaik/OneDrive/Desktop/MLA0405/exp 21.py =====

Warning (from warnings module):
  File "C:/Users/shaik/AppData/Local/Programs/Python/Python313/Lib/site-packages/sklearn/neural_network/multilayer_perceptron.py", line 785
    warnings.warn(
ConvergenceWarning: Stochastic Optimizer: Maximum iterations (1000) reached and the optimization hasn't converged yet.
Accuracy: 0.99
```

